1 Important Constants & Equations

Since we'll be doing a review for chapters 25 to 27 this week, below are all of the equations covered over the last few weeks with only their titles.

Chapter 27

$$i_e = n_e A v_d = \frac{n_e e \tau A}{m} E \qquad \text{(Electron Current in a Wire)}$$

$$I = \frac{dQ}{dt} \approx \frac{Q}{\Delta t} = \frac{eN_e}{\Delta t} = ei_e \qquad \text{(The Current in a Wire)}$$

$$\sigma = \frac{n_e e^2 \tau}{m} \qquad \text{(Conductivity of a Material)}$$

$$I = \frac{\Delta V}{R} \qquad \text{(Ohm's Law)}$$

Chapter 26

$$\Delta V = V_f - V_i = -\int_{s_i}^{s_f} E_s ds = -\int_i^f \vec{E} \cdot d\vec{s} \qquad \text{(The Voltage Difference Generated by an Electric Field)}$$

$$E = -\frac{dV}{ds} \implies E = E_x\hat{\imath} + E_y\hat{\jmath} + E_z\hat{k} = -\left(\frac{\partial V}{\partial x}\hat{\imath} + \frac{\partial V}{\partial y}\hat{\jmath} + \frac{\partial V}{\partial z}\hat{k}\right) \qquad \text{(E-Field of a Potential Diff.)}$$

$$C = \frac{Q}{\Delta V_C} = \frac{\epsilon_0 A}{d} \qquad \text{(Definition of a Capacitor \& its Charge)}$$

$$U_C = \frac{Q^2}{2C} = \frac{1}{2}C(\Delta V_C)^2 \qquad \text{(Potential Energy of a Capacitor)}$$

Chapter 25

$$U_{\text{elec}} = qEs \qquad \text{(Electric Potential Energy in a Uniform Electric Field)}$$

$$V \equiv \frac{U_{\text{q + sources}}}{q} \implies U_{\text{elec}} = qV \qquad \text{(Relationship Between Potential and Potential Energy)}$$

$$E = \frac{\Delta V_C}{d} \qquad \text{(Electric Field of a Parallel-Plate Capacitor)}$$

$$V_{\text{point}} = \frac{1}{4\pi\epsilon_0}\frac{q}{r} \qquad \text{(Electric Potential of a Point Charge)}$$

The following constants/relations are important for the study of electricity:

$e = 1.60 \times 10^{-19}$ C (Fundamental Unit of Charge) $m_p = 1.67 \times 10^{-27}$ kg (Mass of Proton)
$m_e = 9.11 \times 10^{-31}$ kg (Mass of Electron) $K = 8.99 \times 10^{9}$ Nm^2/C^2 (Electrostatic Constant)
$\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 (Permittivity Constant) $\lambda = {dq}/{dl}$ $\eta = {dq}/{dA}$ $\rho = {dq}/{dV}$

There's not much empty space this time :(

2 Problems

1. There is a small square circuit connected to a 5V battery. The sides of the circuit are of length 10cm and the wire has a cross-sectional area of 0.25cm^2. The wire is made out of silver, which has an electron density n_e of 5.8×10^{28} m^{-3}. The circuit has a resistor with an unknown resistance R. Assume the electron drift speed v_d is 1×10^{-5} m/s.

(a) Draw a diagram of this circuit. It may be helpful to draw the cross-sectional area!

(b) What is the electron current i_e of this circuit?

(c) How long would it take for a *single* electron to move through the entire circuit?

(d) What is the current I?

(e) What is the resistance of the unknown resistor R?

2. Suppose there was a conductive wire made of copper with a cross-sectional area 2.5 cm^2 an electron density $n_e = 8.5 \times 10^{28}\ \text{m}^{-3}$, and a conductivity $\sigma = 6.0 \times 10^7\ \Omega^{-1}\text{m}^{-1}$. A current I is passed through the wire with an electron drift speed of $v_d = 5.0 \times 10^-5 {}^{m}/_{s}$.

(a) Find the mean collision time τ.

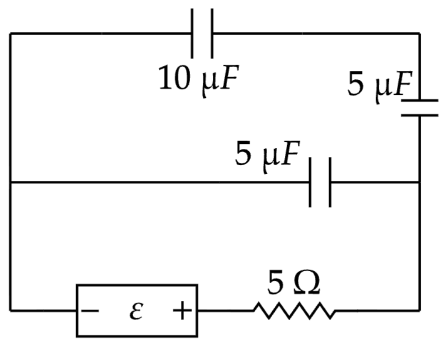
(b) Find the number of electrons passing through a section of the wire in a time period $\Delta t = 5\tau$.

(c) Find the electric field E_{in} inside the wire.

(d) Find the charge located in a small section $L = 10$ cm of the wire. (Hint: Use the electron density!)

(e) Use Gauss's Law to find the electric field created by the small section in part d) a distance r away from the *edge* of the wire.

3. Below is a circuit where $\varepsilon = 5V$. The —/\/\/— symbol represents a *resistor*.



(a) Identity all *sources of charge* and mark them with a ★.

(b) Identify all *sources of voltage* and mark them with a **V**.

(c) Find the equivalent capacitance C_{eq} of this circuit and redraw the circuit.

(d) Analyzing a circuit's response to a capacitor is quite the challenge, but there is a state called the *steady-state* that is much easier to analyze. *(As a side note: This analysis is not actually correct and we should specify we are analyzing the circuit as though the resistor was disconnected for a long time and we re-connected it. The analysis we do here only applies to the very instant we reconnect the resistor. Sorry for the extra long explanation!)*

At the beginning of this steady state, assume that the distance between the plates in the C_{eq} capacitor you found in part (c) is 5cm and the charge *on a single plate* is 5nC. Find the voltage difference ΔV_C across the capacitor.

(e) Now you know the voltage across the battery ε and ΔV_C. Using the value of resistance in the circuit, what is the current I in the beginning of the steady-state?

4. **(CHALLENGE PROBLEM).** You are an evil genius who wants to do something terrible with your lastest innovation, the "ELECTRON" (a particle with a mass of $9.11 \times 10^{-}21$ kg, but with the charge of a regular, boring electron). You want to shoot your friend with an ELECTRON. So you make an ELECTRON gun, but it costs too much to make this prank worthwhile. So you decide to combine the ELECTRON gun with a parallel-plate capacitor of unearthly strength.

   The gun shoots the ELECTRON with an initial speed of $3.00 \times 10^5\,\hat{\imath}\,{}^{m}/_{s}$ and you shoot it through a 3 meter long, 1 meter wide parallel-plate capacitor towards your friend who is 1 meter away from the end of the capacitor. The capacitor has a potential difference of 1 gigavolt and causes an acceleration in the positive-y direction.

   (a) Diagram the scenario, paying special attention to the placement of the potentials, the direction of the resultant electric field E, and an estimated trajectory of the ELECTRON (make sure you show it hitting your friend!). You can ignore any effects of gravity.

   (b) Find the velocity as it hits your friend.

   (c) What would the mass of your friend have to be for the ELECTRON to impart an instantaneous velocity of $v_{\text{friend}} = 1\ {}^{m}/_{s}$ on them?

(d) Conversely, if your friend weighed 100kg, how many ELECTRONs would you need to hit them with at the speed you found in part (b) to impart the same $v_{\text{friend}} = 1\ {}^{m}/_{s}$?

(e) **(CHALLENGE BONUS).** Einstein's Theory of Special Relativity tells us that an object's mass is related to it's energy. Of course, this effect is not noticable since it is nearly impossible to get close to the speed of light. However, because of this mass-energy relationship, it turns out that the mass of a moving object with a rest mass m_0 and velocity u has an inertial mass defined by

$$m = \frac{m_0}{\sqrt{1 - \frac{u^2}{c^2}}},$$

where $c = 3.00 \times 10^8$ is the speed of light.

If we wanted to speed the good ol' electron up to a speed u such that it would have the inertial mass of the rest mass of the ELECTRON (so $m = 9.11 \times 10^-21$ kg), what percent of c would u need to be?